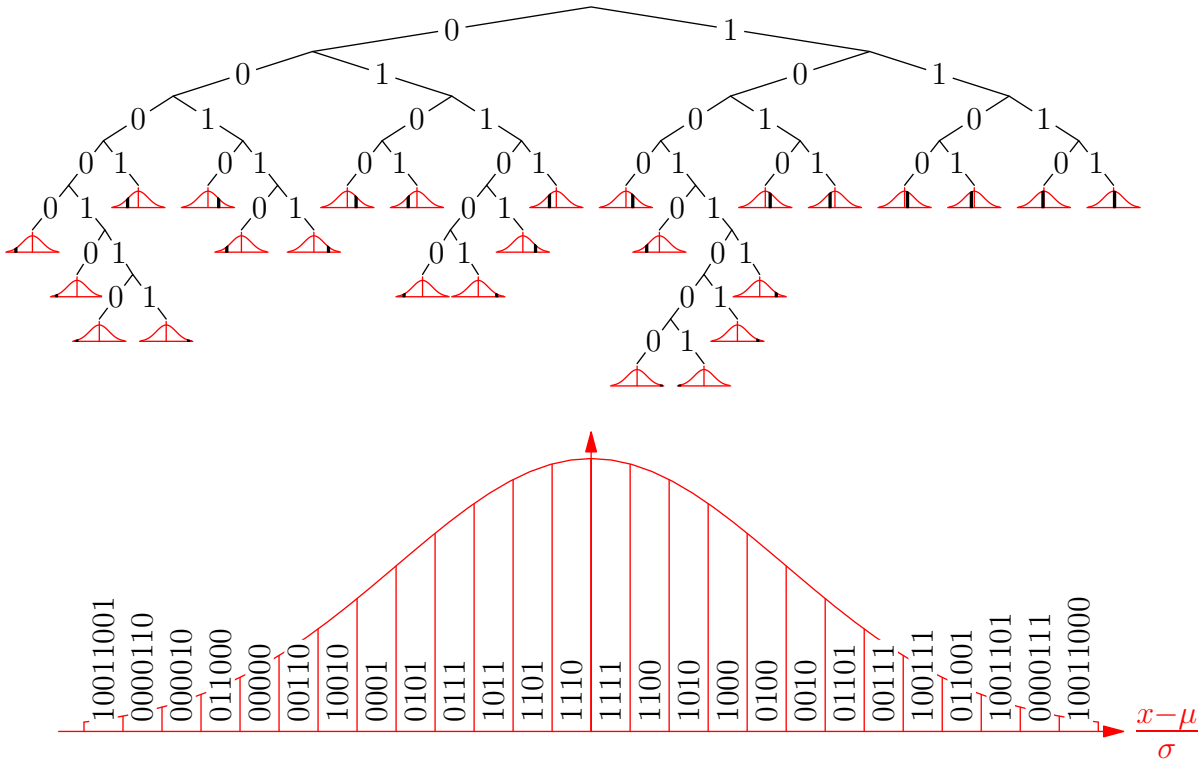
Mathematics of Machine Learning

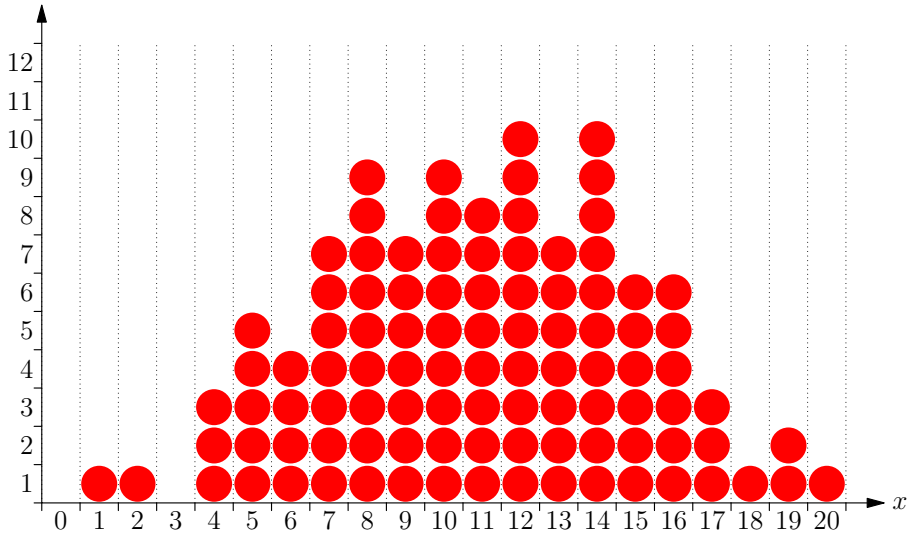
Entropy



Entropy, Coding, Maximum Entropy

Outline

1. **Measuring Uncertainty**
2. Code Length
3. Maximum Entropy



Measuring Uncertainty

- What is more uncertain: tossing a coin three times or throwing a dice?
- The answer depends on whether you care about the order of the coin tosses
- But, how do we answer such a question?
- Let X be a random variable denoting the possible outcomes
- Interestingly, Shannon entropy gives a precise answer

$$H_X = -\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

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Let's Calculate

- For an honest dice $D \in \{1,2,3,4,5,6\}$ and $\mathbb{P}(D=i)=1/6$ so

$$H_D = -\sum_{i=1}^{6} \frac{1}{6}\log_2\left(\frac{1}{6}\right) = -\log_2\left(\frac{1}{6}\right) = \log_2(6) \approx 2.585\text{bits}$$

- For an honest coin where we care about the order so $C \in \{000,001,\ldots,111\}$ the $\mathbb{P}(C=i)=\frac{1}{8}$ and

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- What if we don't care about the order of the outcome? Then $\mathbb{P}(HHH) = \mathbb{P}(TTT) = 1/8$, while two heads (or two tails) in any order have $\mathbb{P}(HHT) = \mathbb{P}(HTT) = 3/8$ so

$$H_U = -\frac{1}{4}\log_2\left(\frac{1}{8}\right) - \frac{3}{4}\log_2\left(\frac{3}{8}\right) \approx 1.811\,\text{bits}$$

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- But why Shannon entropy?

Additive Entropy

- If H_X and H_Y are the uncertainties of two independent random variables X and Y, what is the uncertainty of the combined event (X,Y)?

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H_{(X,Y)} &= -\sum_{X,Y} \mathbb{P}(X,Y)\log_2(\mathbb{P}(X,Y)) \\
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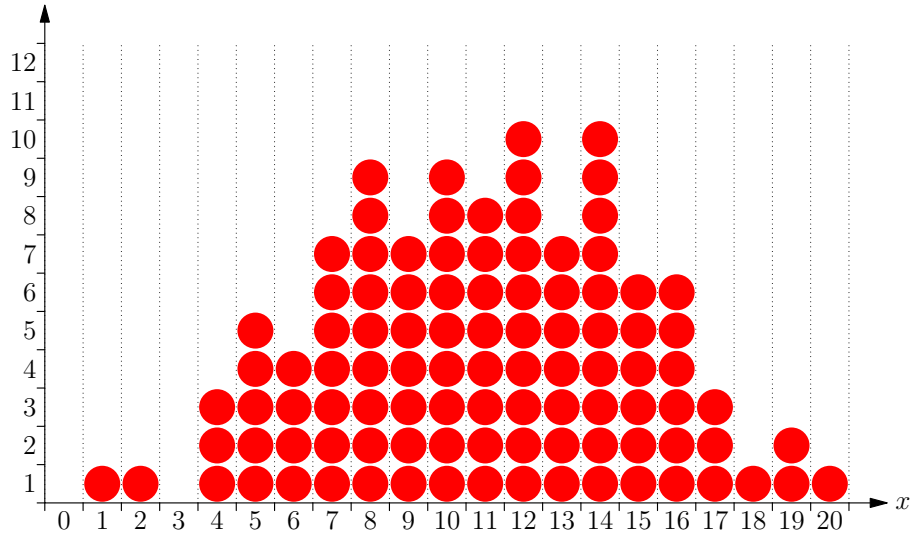
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1. Measuring Uncertainty
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Why Measure Entropy in Bits

- Suppose we had to communicate a message with 2^n equally likely outputs (e.g. the result of n-coin tosses)
- We can do this with a binary string with n bits (011..0)
- If there were 5 possible outcomes I could do this with 3 bits, but waste 3/8 of the message
- However if we have a batch of 3 independent messages each with 5 outcomes then there are 125 possible outcomes
- By batching together enough messages with N outcomes then we asymptotically need just $\log_2(N)$ bits

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Different Probabilities

- We "showed" that if we had N events, X_i, each with probability, $\mathbb{P}(X_i) = 1/N$, we can code the outcomes with a message of length $-\log_2(\mathbb{P}(X_i))$
- With a shorter message we would not be able to distinguish all possible outcomes from the message
- What happens if some of the outcomes occur with a different probability

X_i:	1	2	3	4	5	6
$p(X_i)$:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$
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Shannon's Entropy

- If the probabilities are not equal to 2^{-n} we can still find a code with a length very close to $-\log_2(\mathbb{P}(X))$ per message by transmitting a large number of messages
- The length of the message measures the amount of **surprise** on receiving the message
- Shannon's entropy is the minimum expected length of the message to communicate a random variable X

$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}(X))] = -\sum_{x\in\mathcal{X}} \mathbb{P}(X=x)\log_2(\mathbb{P}(X=x))$$

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- The expected length is a measure of the uncertainty (how much information on average we need to convey the outcome)

Real Codes

- Those of you with some computer science background will realise that with code words of different lengths we must know where each code word ends
- A **prefix code** (no code word is the start of another) can always be decoded
- An optimal prefix code is Huffman encoding where we associate the leaf of a tree with each code word
- Using the tree we can decode any message constructed using the tree
- There is a greedy algorithm for constructing the optimal tree

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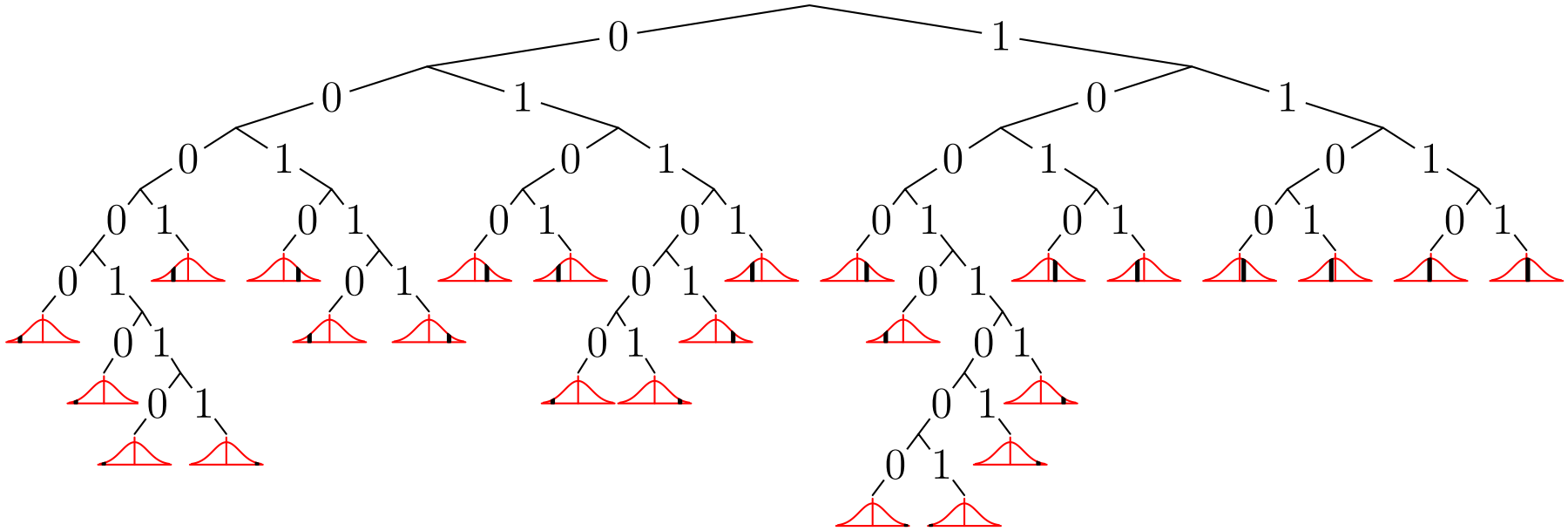
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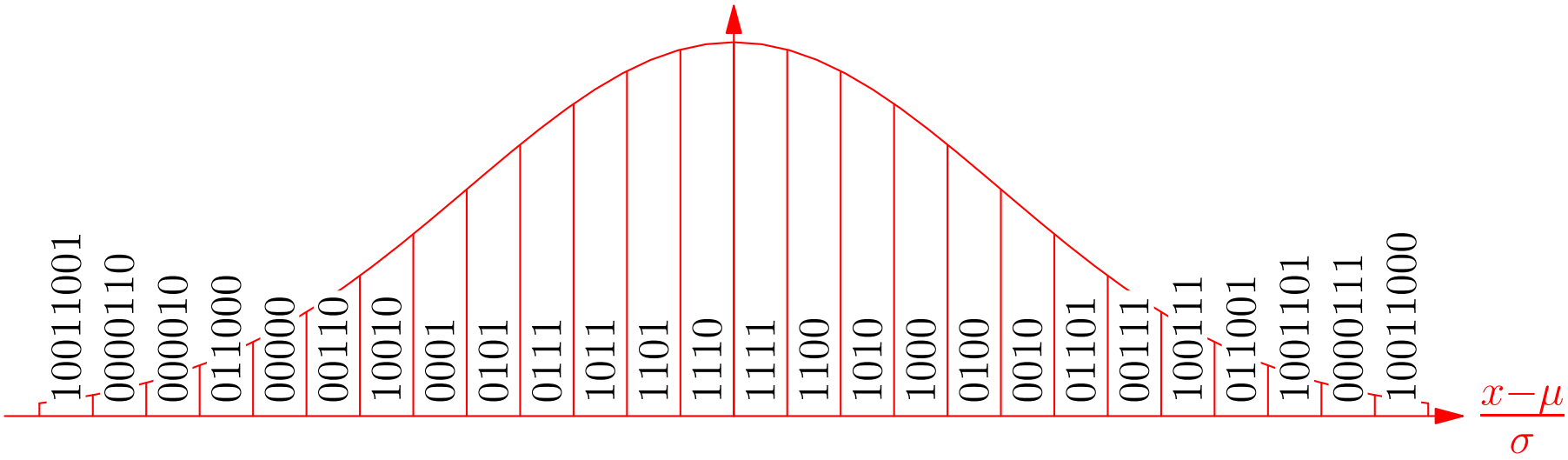
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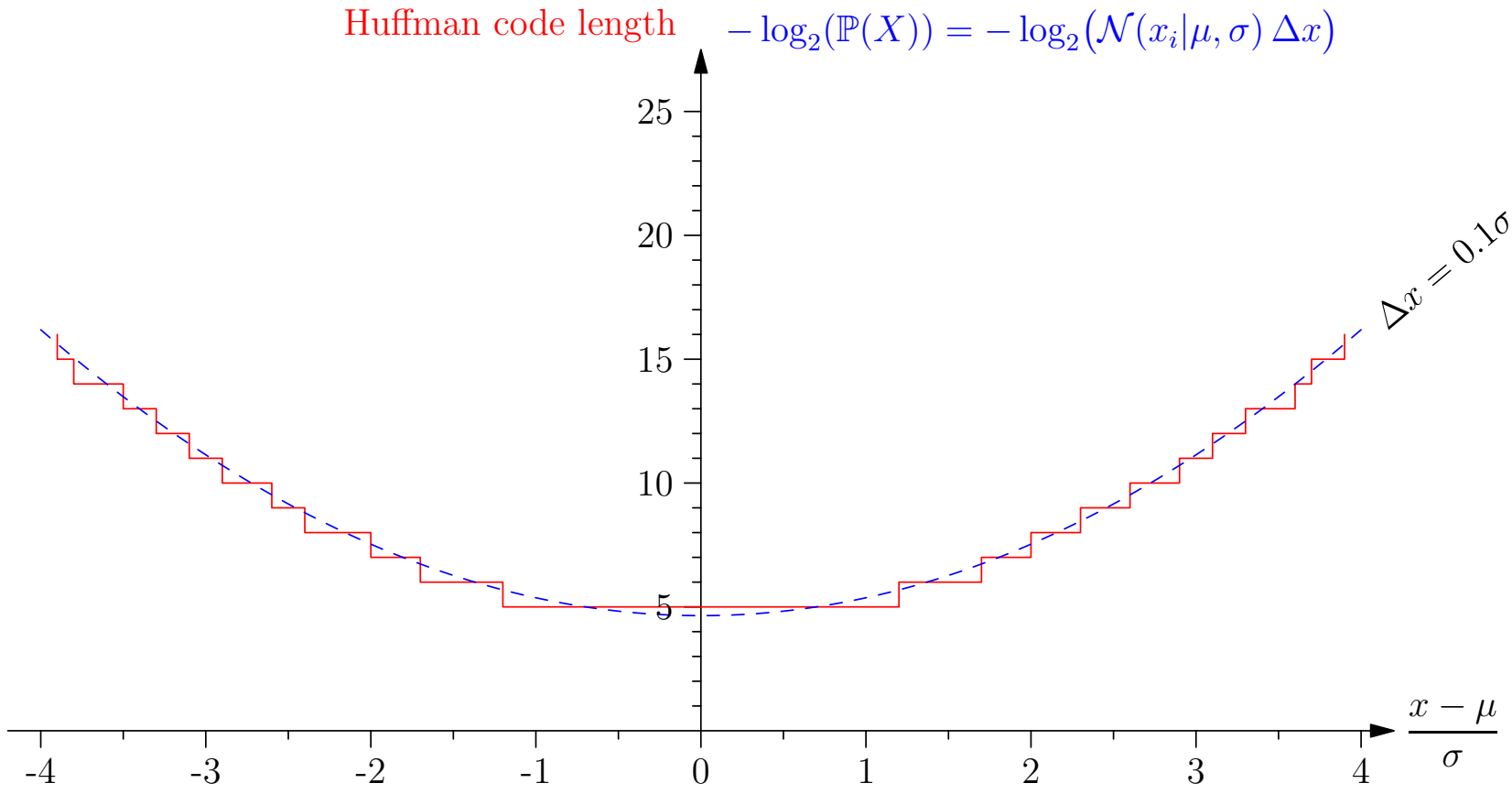
- Those of you with some computer science background will realise that with code words of different lengths we must know where each code word ends
- A **prefix code** (no code word is the start of another) can always be decoded
- An optimal prefix code is Huffman encoding where we associate the leaf of a tree with each code word (its expected length is between H_X and $H_X + 1$ bits)
- Using the tree we can decode any message constructed using the tree
- There is a greedy algorithm for constructing the optimal tree

Coding Normals

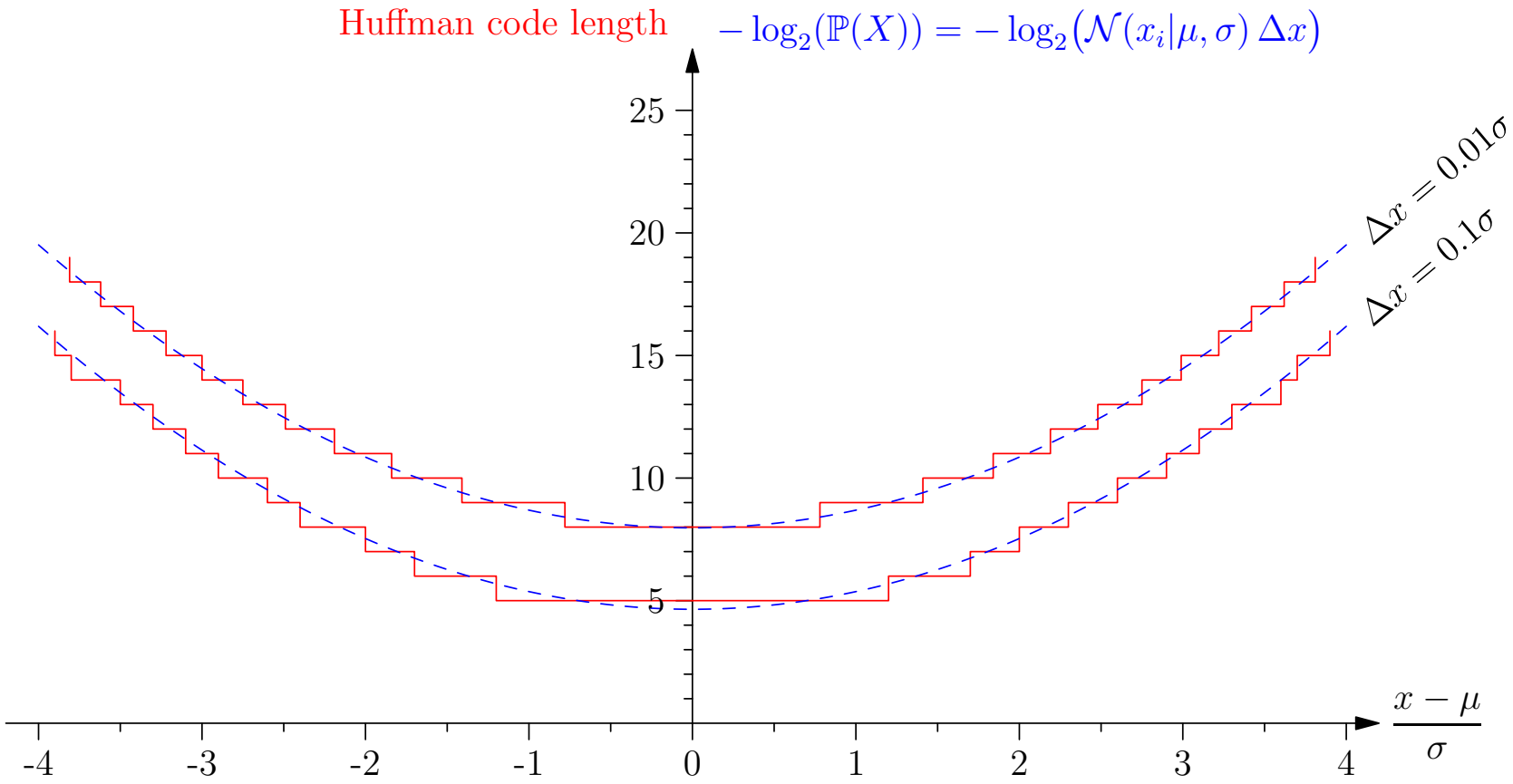




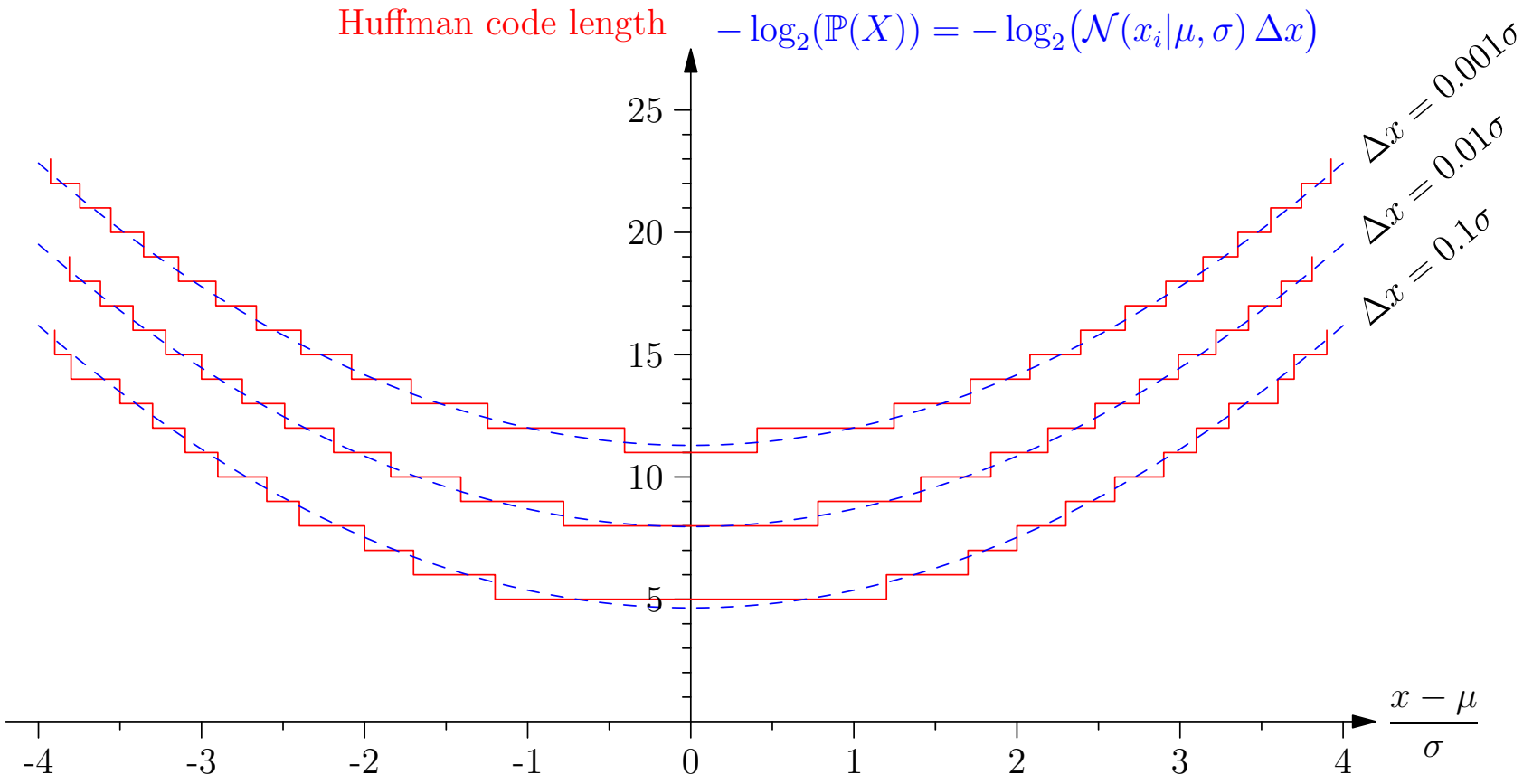
Coding Normals to Accuracy Δx



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bits and nats

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$$H_X = -\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2\big(\mathbb{P}(X = x)\big)$$

- Sometimes it is easier to use natural logarithms

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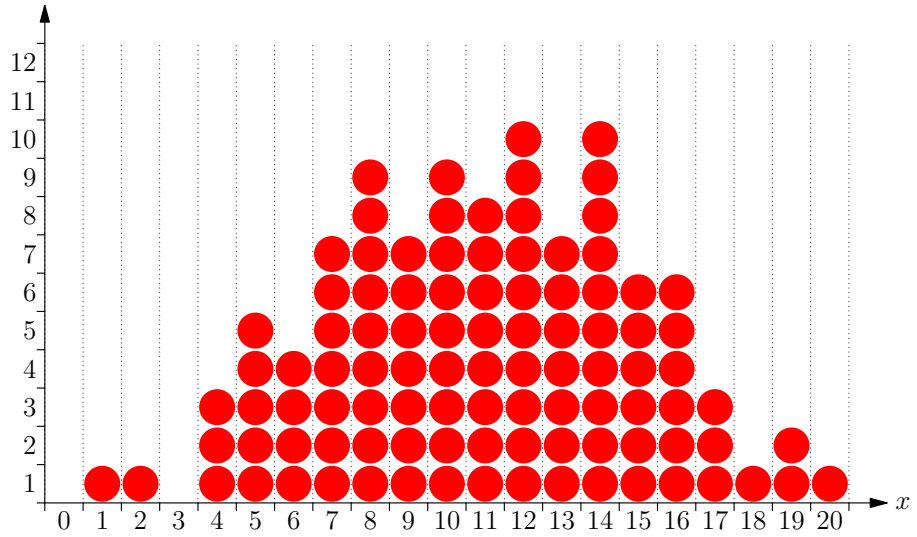
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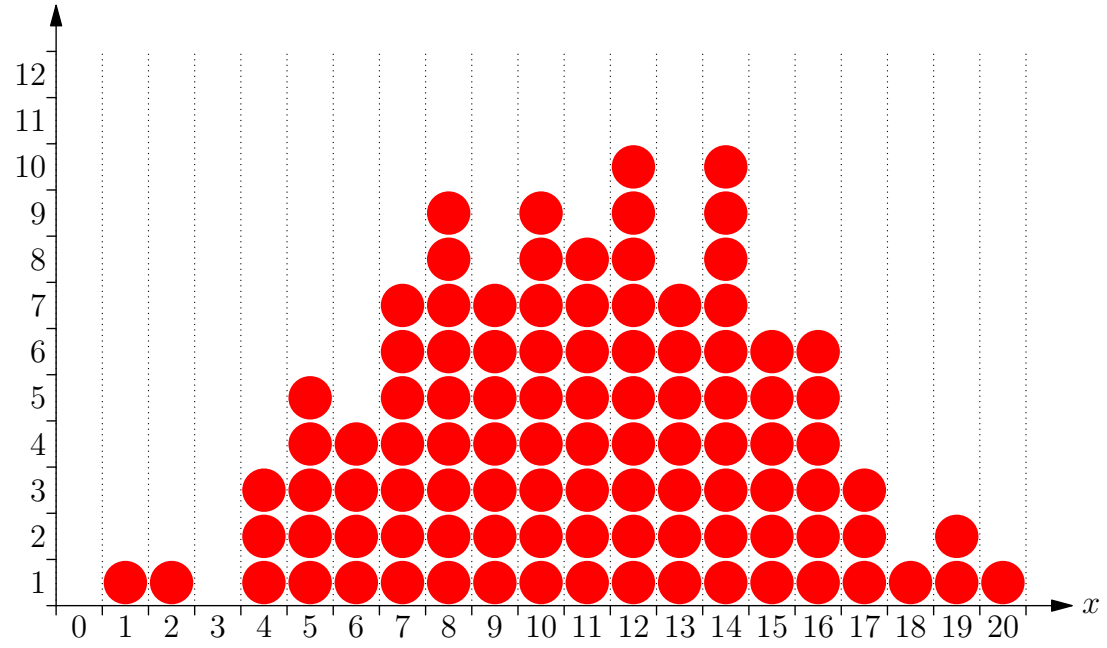
Outline

1. Measuring Uncertainty
2. Code Length
3. **Maximum Entropy**



Number of States

- Suppose I have N balls and put them in K boxes so that n_i balls finish up in box i



- Counting the number of ways to do this

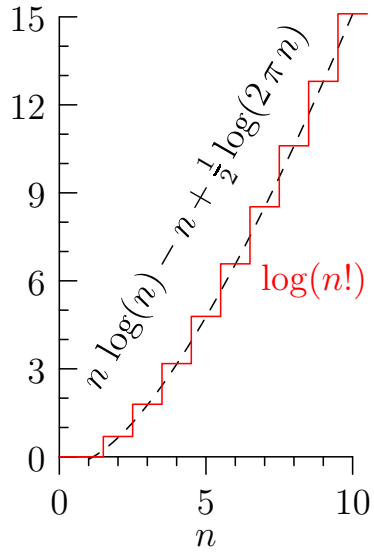
$$W(\boldsymbol{n}) = \frac{N!}{n_1! n_2! \cdots n_K!}$$

Stirling's Approximation

- We can approximate the factorial $n!$ using **Stirling's approximation**

$$n! \approx \sqrt{2\pi n} n^n \mathrm{e}^{-n}$$

$$\log(n!) \approx n\log(n) - n + \frac{1}{2}\log(2\pi n)$$



- Using this in our formula for $W(\boldsymbol{n})$ we have

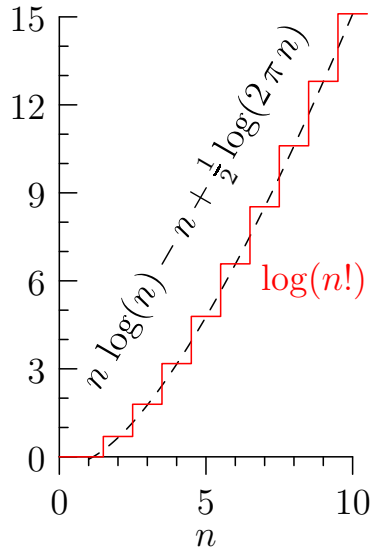
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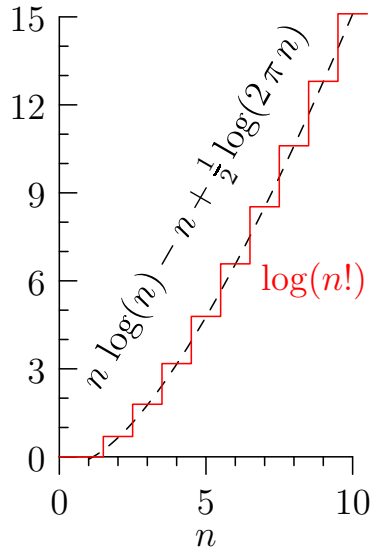
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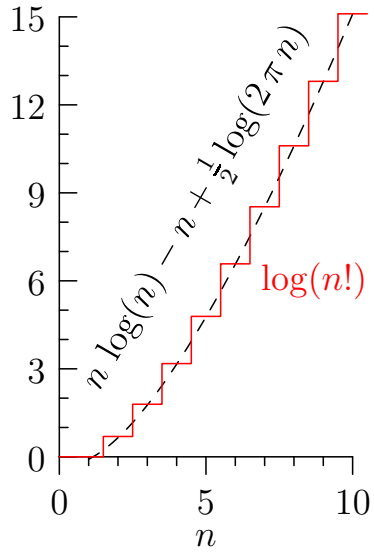
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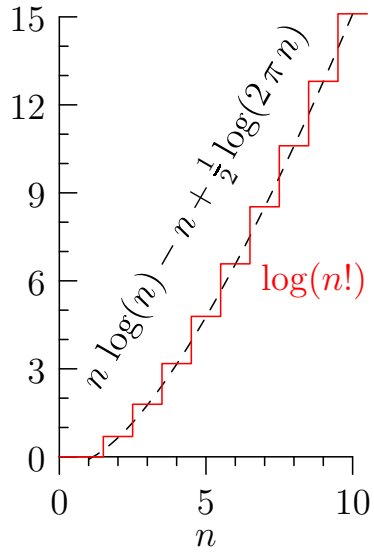
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Number of States and Entropy

- Let $p(x_i) = n_i/N$ be the proportion of balls in bin i then

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$$H_X = -\sum_i p(x_i)\log(p(x_i))$$

- That is, the "entropy" can be seen as a measure of the logarithm of the number of configurations

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Entropy Is Concave

- Each term $h(p) = -p\ln(p)$ has $h''(p) = -1/p < 0$, so the entropy $H = \sum_i h(p_i)$ is a concave function of the probabilities

- Jensen's inequality (convexity lecture) with the concave $\log$ gives, for K outcomes,

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- With linear constraints (fixed averages) the maximum entropy solution is unique
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- When we are trying to infer a distribution given some observations then we can maximise the entropy subject to constraints
- This is known as the **maximum entropy method**
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Knowing the Mean and Variance

- Consider a continuous random variable, X, with a known mean and second moment

$$\mathbb{E}[X] = \mu, \qquad \mathbb{E}\left[X^2\right] = \mu_2 = \mu^2 + \sigma^2$$

- To maximise the entropy subject to constraints consider

$$\mathcal{L}(f) = -\int f_X(x)\log(f_X(x))\mathrm{d}x + \lambda_0\left(\int f_X(x)\mathrm{d}x - 1\right)$$
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- Solving for λ_0, λ_1 and λ_2 then

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

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- A model gives scores s_i to K classes; find the probabilities p_i with maximum entropy and a given average score $\sum_i p_i s_i = \bar{s}$

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- The temperature T is (the inverse of) a Lagrange multiplier, as in statistical physics

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$$p_i = \frac{\mathrm{e}^{s_i/T}}{\sum_j \mathrm{e}^{s_j/T}}, \qquad T = 1/\beta$$

- The temperature T is (the inverse of) a Lagrange multiplier, as in statistical physics

- With scores $s_i = \boldsymbol{w}_i^\mathsf{T} \boldsymbol{x}$ this is multinomial logistic regression, also called the maximum entropy classifier

Softmax Is Maximum Entropy

- A model gives scores s_i to K classes; find the probabilities p_i with maximum entropy and a given average score $\sum_i p_i s_i = \bar{s}$
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Using Maximum Entropy

- Maximum entropy is often used to infer distributions
- It can be very effective, but it might not work well if there are other constraints that we have not included
- The place that they work superbly well is in statistical physics
- The whole of statistical physics is about inferring distributions making observations of volume, pressure, etc.
- Temperature appears rather strangely as a Lagrange multiplier

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Historic Entropy

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- Its interpretation as the number of states was introduced by Ludwig Boltzmann
- The person who got it all right was Josiah Willard Gibbs (and James Clerk Maxwell)
- Claude Shannon invented information theory based on entropy around 1948 (more on that in the next lecture)
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Conclusion

- Entropy provides a measure of the disorder or uncertainty in a system
- It forms the basis of information theory which we will look at in the next lecture
- $-\log(\mathbb{P}(X = x))$ can be seen as the minimum length of a message to communicate x
- This will be used as the basis of the minimum description length formalism also discussed in the next lecture
- Entropy can be used as a prior, which we often maximise subject to constraints to obtain an unbiased estimate

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